

hyper2kvm

Web Dashboard — Product Overview

Browser-based VM migration management with real-time monitoring, multi-provider connectivity, and automated libvirt deployment

h2kweb v0.1.0 — Go + React + Tailwind + WebSocket

<http://185.165.240.5:5070>

Architecture Overview

Full-stack Go backend with React frontend, integrated h2kvmctl CLI execution

Architecture Diagram

```
Browser → React Dashboard (Tailwind CSS, TanStack Query, Zustand)
  ↓ REST API + WebSocket
Go Backend (Chi v5 HTTP Router)
  ├── PAM Authentication (system login)
  ├── Provider Registry (vSphere, Azure, EC2)
  ├── Job Manager (in-memory queue)
  ├── WebSocket Hub (real-time broadcast)
  ├── File Browser API (server filesystem)
  ├── Host Info API (CPU, RAM, disk, libvirt)
  └── h2kvmctl Runner (subprocess + progress parsing)
      ↓ exec.Command + scanLinesOrCR
h2kvmctl (Python CLI) → Convert → Fix → Deploy → virsh define+start
```



Go Backend

20MB static binary, Chi v5 router,
WebSocket, PAM auth, systemd service



React Frontend

Tailwind CSS (v9s design), TanStack
Query, Zustand, Lucide icons, 87KB
gzipped

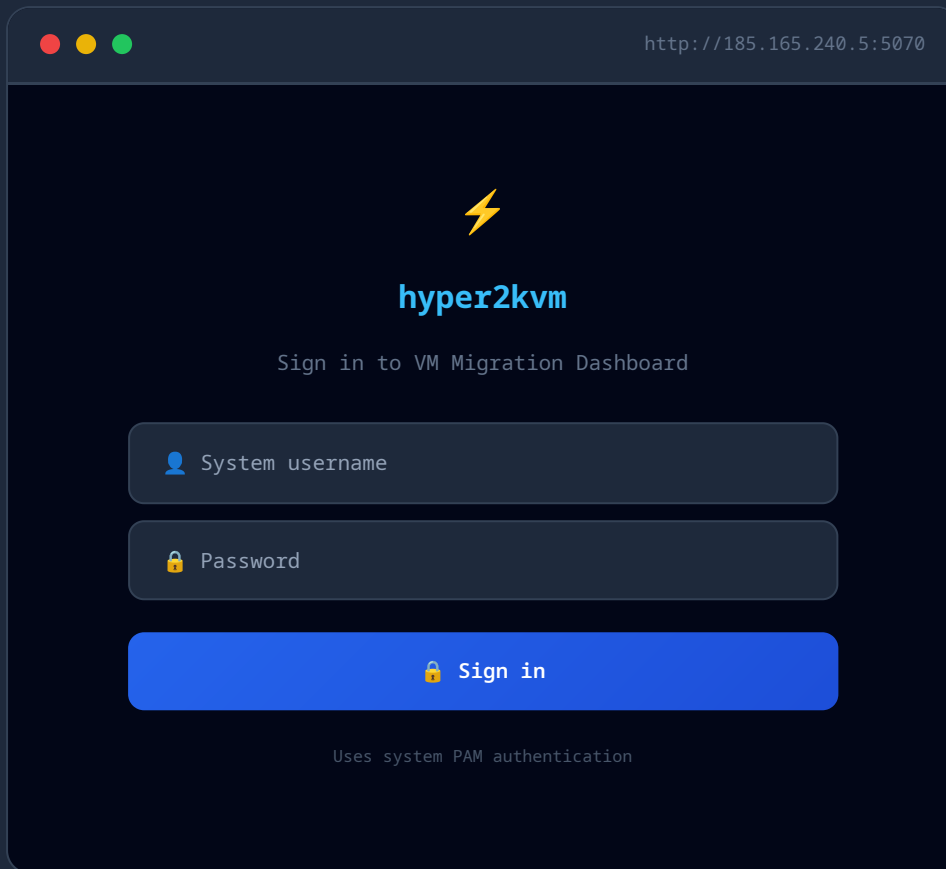


h2kvmctl Integration

Subprocess execution, YAML config
generation, live stdout parsing,
progress tracking

PAM Authentication

System-level authentication using Linux PAM — same credentials as SSH



A screenshot of a web browser window showing the login page for 'hyper2kvm'. The browser's address bar displays 'http://185.165.240.5:5070'. The page has a dark background. At the top center is a yellow lightning bolt icon, followed by the text 'hyper2kvm' in a light blue font. Below this is the text 'Sign in to VM Migration Dashboard'. There are two input fields: the first is labeled 'System username' with a person icon, and the second is labeled 'Password' with a lock icon. Below these fields is a large blue button with a lock icon and the text 'Sign in'. At the bottom of the page, it says 'Uses system PAM authentication'.

Security Features

- ✓ PAM authentication against system users
- ✓ HttpOnly session cookies (SameSite=Strict)
- ✓ 32-byte random hex session tokens
- ✓ In-memory session store (no external DB)
- ✓ API key fallback for CLI/scripts
- ✓ Public endpoints: /health, /auth/* only
- ✓ All API routes require valid session
- ✓ Username character validation
- ✓ CORS middleware for browser access

Auth Flow

```
POST /api/v1/auth/login
→ PAM verify (pam.StartFunc "login")
→ Create session (32-byte token)
→ Set-Cookie: h2kweb_session=<token>
```


Dashboard — Real-time Overview

Live host metrics, migration pipeline status, provider health, and activity feed

8

CPU Cores

15.6 GB

Memory (43% used)

219 GB

Disk (90% used)

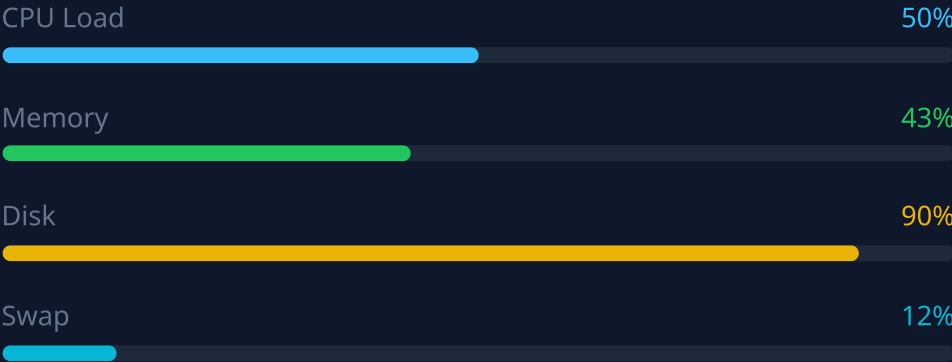
3

Libvirt VMs

Host Information

Hostname	NLDW3-4-31-12
OS	AlmaLinux 9.7 (Moss Jungle Cat)
Kernel	5.14.0-611.41.1.el9_7.x86_64
CPU	Intel Xeon E3-1230 v6 @ 3.50GHz
QEMU	9.1.0
KVM	✓ Enabled
Uptime	3d 4h 59m
Load	4.04 / 4.16 / 4.42

Resource Utilization



Libvirt Domains

ubuntu2404 • running

centos10 • running

esxi-lab • shut off

Providers — Multi-Cloud Connectivity

Connect to VMware vSphere, Microsoft Azure, and AWS EC2 with dedicated login cards



VMware vSphere

Connect to vCenter Server via govmomi SDK. Browse VMs, discover disks, export and migrate.

vCenter Host · Username · Password ·
Datacenter



Microsoft Azure

Connect via Azure CLI. List VMs by subscription and resource group, download managed disks.

Subscription ID · Resource Group



AWS EC2

Connect via AWS CLI. Discover EC2 instances, export AMIs, migrate to KVM with full fixes.

Region · AWS Profile

After Connecting — VM Browser

Provider → VM Browser

Search VMs...

☒ web-server-01	4 vCPU · 8192MB · 120GB	● poweredOn
☒ db-server-02	8 vCPU · 16384MB · 500GB	● poweredOn
☐ dev-sandbox	2 vCPU · 4096MB · 60GB	● poweredOff
☐ legacy-app	4 vCPU · 8192MB · 200GB	● poweredOn

Migration Wizard

4 source types, server file browser, full YAML configuration, and YAML preview



Local File

Browse server filesystem for
VMDK, OVA, VHD



VMware vSphere

Select from connected
vCenter



Microsoft Azure

Select from Azure
subscription



AWS EC2

Select EC2 instances

Server File Browser

🔴 🟡 🟢 / root / demo

3 dirs 4 VM files

▢ Show all files

📁 vmware-exports/

🔴 centos10.vmdk

VMDK

2.2 GB

🔴 centos9.vmdk

VMDK

1.8 GB

🔴 fedora42.vmdk

VMDK

4.6 GB

✓ Selected

🔴 ubuntu.ova

OVA

3.1 GB

🔴 Selected: /root/demo/fedora42.vmdk

VMDK

Configuration Options (All YAML)

▼ Output

Output Directory · Format (qcow2/raw/vdi) · Filename

▼ Processing

☒ Flatten Snapshots · ☒ Compress Output · Level

▼ Guest Fixes

fstab Mode · ☒ Regen Initramfs · ☒ Update GRUB · ☒ Remove VMware Tools

▼ Libvirt / Deploy

☒ Emit XML · ☒ virsh define · ☒ Start VM · ☒ Keep Domain · Name/Memory/vCPUs

▶ Kubernetes

Deploy to KubeVirt · Namespace

Job Monitor — Live Streaming

Real-time h2kvmctl output via WebSocket, deployment status, IP detection

Live Output (WebSocket)

Terminal • streaming 307 lines

```
1  ✓ Loading config: /tmp/h2kweb-xxx.yaml
2  ✓ Sanity: OK
3  ✓ Mode: local
4  ✓ Input VMDK: centos10.vmdk (2.20 GiB)
5  ✓ Quick probe: detected LVM on disk
6  ✓ Flattened: working-flattened.qcow2
7  ✓ VMCraft ready in 3.38s (1 drive)
8  ✓ LVM: cs-root, cs-home, cs-swap
9  ✓ Root: /dev/mapper/cs-root (xfs)
10  🛠 Running: dracut -f --add-drivers virtio...
11  ✓ Completed in 26.3s
12  ✓ grub2-mkconfig: Succeeded
13  ✓ Conversion complete: centos10.qcow2
14  ✓ Domain: centos10 → RUNNING
```

Deployment Status



Libvirt

Domain centos10

State **running**



Network

● 192.168.122.45

📄 SSH

Job Card States

⌚ pending

Queued, waiting to start

🔄 running

Converting, with progress bar

✓ completed

Done, shows deployment status + IP

✗ failed

Error message displayed

REST API Reference

12+ endpoints with PAM authentication, JSON responses, WebSocket streaming

Method	Endpoint	Description
POST	/auth/login	PAM login
POST	/auth/logout	Destroy session
GET	/auth/session	Check session
GET	/health	Health check
GET	/status	Job stats + providers
GET	/host/info	CPU/RAM/disk/libvirt
POST	/providers	Add provider
GET	/providers/{id}/vms	List VMs
GET	/browse?path=	Browse filesystem
POST	/migrations/submit	Start migration
GET	/jobs	List all jobs
GET	/deploy/{vm}/ip	VM IP detection

Deployment

```
# Install (one command)
$ cd web && sudo make install

# Service management
$ systemctl status h2kweb
$ journalctl -u h2kweb -f

# Remote deploy
$ ./scripts/deploy-remote.sh 10.0.0.1 root

# Config: /etc/default/h2kweb
H2KWEB_ADDR=:5070
```

Tech Stack

Go 1.23

Chi v5

React 18

Tailwind 3

TanStack Query 5

Zustand

WebSocket

PAM

govmomi

Lucide Icons

Binary Size

Go backend	20 MB (static binary)
React frontend	88 KB gzipped

WS	/ws	Real-time events	CSS (Tailwind)	6 KB gzipped
			systemd service	Built-in